

# AUTO CLASSIFICATION USING FLOW DESIGNER

## DESIGN PHASE

| TEAM 10       | NAME              | NM ID               |
|---------------|-------------------|---------------------|
| TEAM LEADER   | JEEVITHA. M       | asaluau020224128017 |
| TEAM MEMBER 1 | JAMUNA. S         | asaluau020224128016 |
| TEAM MEMBER 2 | GUNA SUNDARI. T   | asaluau020224128015 |
| TEAM MEMBER 3 | DIVYA BHARATHI. B | asaluau020224128013 |
| TEAM MEMBER 4 | PRIYA MANI. J     | asaluau020224128027 |

### 1. Introduction:

- The Design Phase is an important stage in the development of the Auto Ticket Classification using Flow Designer project. The main purpose of this phase is to design the structure, fields, workflow, automation logic, and user interaction of the system before implementing the complete solution.
- The design is created based on the requirements identified during the Requirement Analysis and Planning phase. The system is designed to automatically classify IT support tickets based on the issue description provided by the user.
- The major issue categories considered in the project are Network, Hardware, Access, and Performance. Each category is associated with a suitable Subcategory.

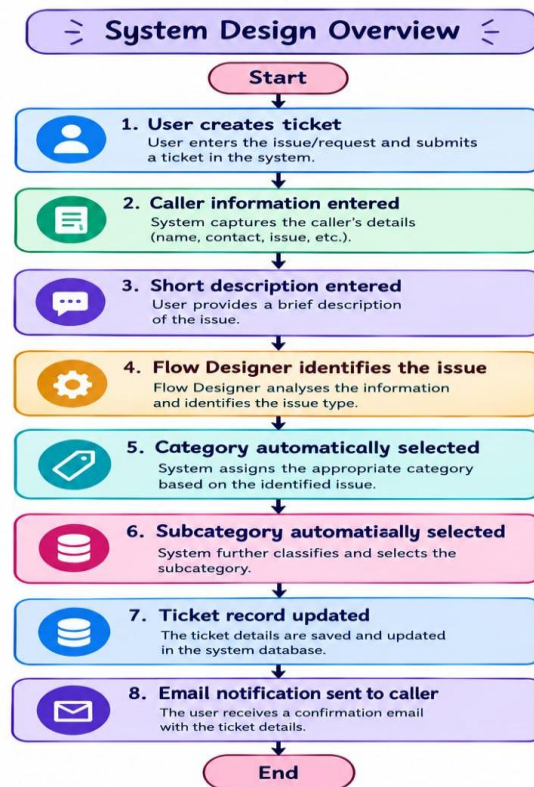
### 2. Objectives of the Design Phase:

The main objectives of the Design Phase are:

- To design the overall structure of the ticket management system.
- To identify and define all required fields.
- To select suitable data types for each field.
- To design Category and Subcategory choices.
- To establish dependency between Category and Subcategory.
- To design the ticket classification workflow.

### 3. System Design Overview:

The Auto Ticket Classification system is designed as a structured IT helpdesk ticket management system.



The basic system

- This design reduces the need for manual ticket classification and provides a standardized approach for handling IT support requests.

#### 4. Database and Table Design:

- A custom table named Incident Workflow is designed to store the IT support ticket records.
- The table is designed with the required fields for storing ticket information in a structured format.

Table Name

Incident Workflow

The table contains fields such as:

|   |                   |
|---|-------------------|
| 1 | Number            |
| 2 | Caller            |
| 3 | Category          |
| 4 | Subcategory       |
| 5 | Short Description |
| 6 | Description       |

|   |                |
|---|----------------|
| 7 | State          |
| 8 | Assigned Group |
| 9 | Assigned to    |

- The Number field is designed as an Auto Number field so that each ticket can be identified using a unique ticket number.
- The Caller field is designed as a Reference field to the user table. Similarly, Assigned Group and Assigned To are reference fields used for assigning and managing tickets.

## 5. Field Design:

- The following fields are designed for the Incident Workflow table:

| FIELD LABEL       | TYPE        | PURPOSE                                      |
|-------------------|-------------|----------------------------------------------|
| Number            | Auto Number | Unique ticket identification.                |
| Caller            | Reference   | Stores the user who creates the ticket.      |
| Category          | Choice      | Identifies the main issue type.              |
| Subcategory       | Choice      | Identifies the specific issue.               |
| Short Description | String      | Stores a short description of the issue.     |
| Description       | String      | Stores detailed information about the issue. |
| State             | Choice      | Tracks the ticket status.                    |
| Assigned Group    | Reference   | Identifies the responsible group.            |
| Assigned To       | Reference   | Identifies the assigned user.                |

The reference project defines these fields with appropriate data types, including Auto Number, Reference, Choice, and String fields.

## 6. Category Design:

The Category field is designed as a Choice field.

The following categories are included:

1. Network

## 2. Hardware

## 3. Access

## 4. Performance

- Each category represents a major type of IT support problem.
- Network
- This category is used for network-related problems such as Wi-Fi connectivity issues.
- Hardware
- This category is used for hardware-related issues such as projector failures.
- Access

This category is used for account access problems such as forgotten passwords.

## Performance

This category is used for computer performance problems such as slow or hanging computers.

## 7. Subcategory Design:

- The Subcategory field is also designed as a Choice field.
- The following Subcategories are included:

- Wi-Fi

- Projector

- Forgot Password

- Slow Computer

Each Subcategory is associated with a particular Category.

The design is structured as follows:

Network → Wi-Fi

Hardware → Projector

Access → Forgot Password

Performance → Slow Computer

This structure makes ticket classification more organized and consistent.

## 8. Dependent Field Design:

- The Category and Subcategory fields are designed using a dependent relationship.
- The purpose of this design is to make the Subcategory options depend on the selected Category.
- For example, when the user selects Network as the Category, the appropriate Subcategory such as Wi-Fi can be displayed.

Similarly:

- Network → Wi-Fi
- Hardware → Projector
- Access → Forgot Password
- Performance → Slow Computer

This design prevents unrelated Subcategory values from being selected.

## 9. Steps to Configure Dependent Fields:

The dependency between Category and Subcategory is designed using the dictionary configuration.

Navigation Steps:

1. Open the created Incident Workflow table.
2. Open the table form.
3. Right-click on the Subcategory field.
4. Select Configure Dictionary.
5. Open the Advanced View.
6. Locate the Dependent Field section.
7. Enable Use dependent field.
8. Select Category in the Dependent on field.

## 9. Save the configuration.

After configuring the dependency, the Subcategory values are displayed based on the selected Category.

The reference document uses the same dependency design and maps Wi-Fi to Network, Projector to Hardware, Forgot Password to Access, and Slow Computer to Performance.

## 10. State Field Design:

The State field is designed as a Choice field to track the current status of the ticket.

The following choices are included:

- New
- In Progress
- On Hold
- Resolved
- Closed

The State field allows IT staff to understand the current stage of each ticket.

For example:

New – Ticket has been created.

In Progress – Ticket is currently being worked on.

On Hold – Work on the ticket is temporarily stopped.

Resolved – The reported issue has been resolved.

Closed – The ticket has been completely closed.

## 11. Form Design:

The Incident Workflow form is designed to provide a simple and structured interface for entering ticket information.

The form contains the following important fields:

- Caller

- Short Description
- Description
- Category
- Subcategory
- State
- Assigned Group
- Assigned To

The form is designed so that users can enter the required information easily.

The Short Description field plays an important role in the automation because the Flow Designer uses the information entered in this field to identify the type of issue.

## 12. User Interface Design:

- The user interface is designed to make ticket creation simple for students and teachers.
- The user needs to provide basic information about the issue. The system then processes the
- Information and automatically performs classification.

The design focuses on:

- Simple ticket creation.
- Clear field labels.
- Structured choice values.
- Easy selection of required information.
- Reduced manual classification.
- Consistent ticket information.

This design helps reduce unnecessary manual work for IT staff.

## 13. Flow Design:

The Flow Designer automation is designed to process newly created tickets.

The flow is named:

Auto Classify School IT Tickets

- The flow is designed to run when a new record is created in the Incident Workflow table.
- The trigger is:
- Record Created
- The flow checks whether the Category field is empty before performing automatic classification.
- The reference project uses the Flow Designer trigger Record Created with the Incident Workflow table and a condition where Category is empty.

#### 14. Automatic Classification Logic Design:

- The main design component of this project is the automatic classification logic.
- The system is designed to analyse the Short Description entered by the user.
- Based on predefined keywords, the system identifies the issue type and updates the Category and Subcategory fields.

The classification design is as follows:

Wi-Fi Issue

If the Short Description contains Wi-Fi or Network:

Category → Network

Subcategory → Wi-Fi

Projector Issue

If the Short Description contains Projector:

Category → Hardware

Subcategory → Projector

Password Issue

If the Short Description contains Password or Login:

Category → Access

Subcategory → Forgot Password

Slow Computer Issue

If the Short Description contains Slow or Hanging:

Category → Performance

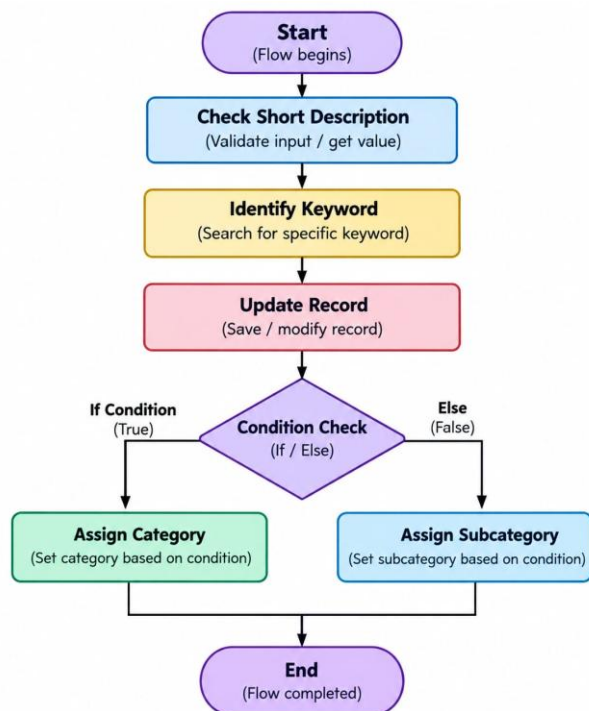
Subcategory → Slow Computer

The reference design specifies these keyword-based classification rules within Flow Designer.

## 15. Flow Logic Design:

The Flow Designer is designed using conditional logic.

The basic structure is:



The flow uses If and Else If conditions to check different types of IT issues.

For example:

If Short Description contains Wi-Fi

→ Category = Network

→ Subcategory = Wi-Fi

Else If Short Description contains Projector

→ Category = Hardware

→ Subcategory = Projector

Else If Short Description contains Forgot Password

→ Category = Access

→ Subcategory = Forgot Password

Else If Short Description contains Slow Computer

→ Category = Performance

→ Subcategory = Slow Computer

## 16. Update Record Design:

After identifying the issue, the Flow Designer is designed to use the Update Record action.

The Update Record action updates the ticket with the appropriate Category and Subcategory.

For example:

Wi-Fi

Category = Network

Subcategory = Wi-Fi

Projector

Category = Hardware

Subcategory = Projector

Password

Category = Access

Subcategory = Forgot Password

Slow Computer

Category = Performance

Subcategory = Slow Computer

The reference flow uses Update Record actions to automatically assign these values.

## 17. Email Notification Design:

- An automated email notification is included in the system design.
- After the ticket is classified successfully, the Flow Designer sends an email notification to the caller.
- The email design contains:

To: Caller Email

Subject: Your Request for the issue has been submitted.

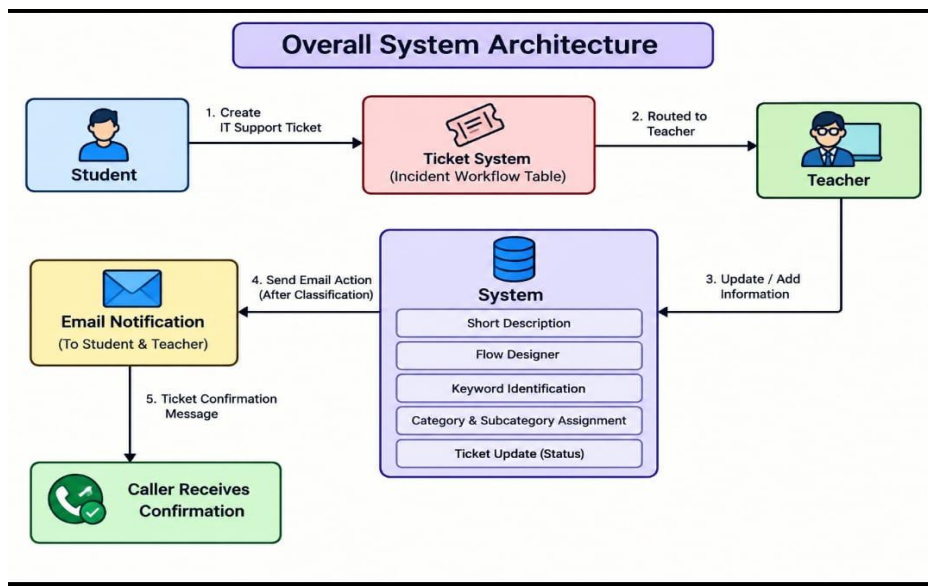
Body: Ticket confirmation message.

This notification informs the caller that the IT support request has been successfully submitted.

The reference project includes a Send Email action after classification, with the caller's email used as the recipient.

## 18. Overall System Architecture:

The overall design of the system can be represented as follows:



This architecture provides a simple no-code automation structure for the school IT helpdesk.

## 19. Design Considerations:

- The following points are considered while designing the system:
  - The system should be easy to understand.
  - The fields should have appropriate data types.
  - Category and Subcategory should be properly connected.
  - The automation should work without manual classification.
  - The design should support future modifications.

## 20. Complete Design Workflow:

- The complete design of the Auto Ticket Classification system follows a structured workflow.
- Step 1 – Ticket Creation
  - The student or teacher creates an IT support ticket by entering the Caller information and Short Description.
- Step 2 – Record Creation
  - The ticket is stored in the Incident Workflow table.
- Step 3 – Flow Trigger
  - The Flow Designer is triggered automatically when the record is created.
- Step 4 – Condition Checking

- The flow checks whether the Category field is empty.
- Step 5 – Keyword Identification

The Short Description is checked for predefined keywords.

#### Step 6 – Classification

- The appropriate Category and Subcategory are selected automatically.
- Step 7 – Record Update
- The ticket record is updated with the classification information.
- Step 8 – Email Notification
- An automated confirmation email is sent to the caller.

### 21. Design Mapping:

- Issue Type| Keyword| Category| Subcategory
- Wi-Fi Issue| Wi-Fi / Network| Network| Wi-Fi
- Projector Issue| Projector| Hardware| Projector
- Password Issue| Password / Login| Access| Forgot Password
- Slow Computer| Slow / Hanging| Performance| Slow Computer
- This mapping provides a clear relationship between the issue description and the classification values.

### 22. Benefits of the Designed System:

- The design provides several benefits to the school IT helpdesk.
- Reduced Manual Work
- IT staff do not need to manually identify the category for every ticket.
- Consistent Classification
- The same keywords always lead to the predefined Category and Subcategory values.
- Faster Ticket Processing
- Automatic classification allows tickets to be processed quickly.
- Structured Data
- All ticket information is stored in a standardized format.
- Easy Maintenance
- The system is developed using Flow Designer and does not require complex scripting.

### 23. Design Summary:

- ❖ The Design Phase establishes the complete structure of the Auto Ticket Classification using Flow Designer system.
- ❖ The system design includes the Incident Workflow table, required fields, Category and Subcategory choices, dependent field configuration, State field, Flow Designer logic, automatic classification, and email notification.
- ❖ The design follows a structured approach where the user's Short Description is used as the primary input for automatic classification.
- ❖ The Flow Designer automation is designed to identify predefined keywords and automatically update the ticket record. After classification, an email notification is sent to the caller.

## 24. Conclusion:

- ❖ The Design Phase provides a clear blueprint for implementing the Auto Ticket Classification system. It defines how ticket information is stored, how fields are related, how issue classification is performed, and how notifications are generated.
- ❖ . The completed design provides a structured, simple, and maintainable foundation for the next development and testing activities. By using Service Now Flow Designer, the system can perform automatic ticket classification without requiring complex programming or scripting.
- ❖ Therefore, the Design Phase successfully establishes the technical and functional structure required for the implementation of the Auto Ticket Classification using Flow Designer project.